PROJECT-BASED LEARNING

PHASE-I EVALUATION

Ambulance and Emergency Dispatch System

Team ID: DSCPP-III-2026-T148

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T148
Semester: 3rd
Section: K, E, ML1, ML3
Domain: *B.tech CSE*

Team Members

1. *Neel Hajariwala – 2510011575*
2. *Keshav Namdev – 2510360097*
3. *Lichhavi Chauhan – 2510370231*
4. *Aditya Ahlawat – 2512510001*

Mentor

Mr.Ramesh Singh Rawat

Interactions completed: ______

Problem Statement & Motivation

Problem Statement

In emergencies, manual ambulance dispatch can cause delays in prioritizing cases and assigning the nearest available ambulance. This project addresses this gap by developing a C++ and Data Structures-based system that efficiently prioritizes emergency requests and dispatches suitable ambulances, improving response time and resource utilization.

Motivation

- *Delays*
- *Inefficacy*
- *Mortality*

Target Users / Stakeholders

- *Dispatchers*
- *Hospitals*
- *Efficiency*

Objectives

Project Objectives

1. *Develop: C++ dispatch system.*
2. *Implement: Priority queues and graph algorithms.*
3. *Integrate: DSA and OOP principles.*
4. *Evaluate: Response times and queue speeds.*
5. *Demonstrate: Scalable real-time ambulance routing.*

Key Objective: *Minimize dispatch delay via automated priority routing.*

Proposed Solution

Proposed Approach

1. *Input: Call data.*
2. *Processing: Priority routing.*
3. *Logic: Auto allocation.*
4. *Database: Status logs.*
5. *Output: Dispatch details.*

Key Features

- *Triage: Max-Heap queues.*
- *Routing: Dijkstra paths.*
- *Tracking: Live monitoring.*
- *Speed: Zero delays.*
- *Impact: Faster care.*

Automates patient triage using Max-Heap priority queues and optimizes ambulance navigation via Dijkstra's shortest-path algorithm, directly eliminating manual dispatch delays and minimizing emergency response times.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, Linked Lists, Trees, Graphs, Hashing, etc.*	*Triage Engine*
OOPs with C++	*Classes, Inheritance, Polymorphism, STL, etc.*	*Dispatch Manager*

Requirement: Integrates dynamic C++ data structures with automated system architecture.

System Architecture / Workflow

Major Components

- *Triage Engine*
- *Routing Module*
- *Database Layer*

Data / Control Flow

- *Sequential*
- *Automated*
- *Assignment*

Technology Stack & Methodology

Technology Stack

- Programming: *C++*
- Tools / Framework: *STL*
- IDE: *VS Code*
- Version Control: *GitHub*

Development Methodology

1. Emergency Triage Scoping.
2. Heap logic.
3. C++ Core Coding.
4. Dijkstra Path Validation.
5. Performance optimization.

C++ delivers high-performance execution for instant pathfinding, file-based logging ensures lightweight zero-overhead storage, and Agile methodology enables rapid, iterative testing of critical emergency logic.

Initial Progress & Team Contribution

Progress Achieved

- *Scoping*
- *Design*
- *Prototyping*
- *Structuring*
- *Datasets*

Individual Contribution

Member	Role	Contribution
Neel Hajariwala	*Lead*	*Project planning, system architecture, coordination and module integration*
Keshav Namdev	*Developer*	*Max-Heap/Priority Queue, Dijkstra algorithm and C++ implementation*
Lichhavi Chauhan	*Developer/Docs*	*Workflow, flowchart, module development and documentation*
Aditya Ahlawat	*DB/Testing*	*Data management, ambulance status/logs, testing and validation*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Architecture & Triage Design
2. Phase-II: Graph Routing & Heap Coding
3. Phase-III: System Testing & Integration

Expected Outcomes

- *Working C++ dispatch prototype*
- *Reduced emergency response times*
- *Functional shortest-path routing*
- *Scalable multi-ambulance queue*

References

1. *Journal – Introduction to Algorithms*
2. *https://www.youtube.com/watch?v=l_xg60cFvX0*
3. *Map and Road Network Data*
4. *GeeksforGeeks – Dijkstra's Algorithm and Priority Queue*

Thank You
Questions & Suggestions